Lab 01

Number Systems

Task 01: Decimal Conversions

Convert the following decimals numbers into **Binary**, **Octal**, and **Hexadecimal**:

1. 25

Binary

2	25	
2	12	- 1
2	6	- 0
2	3	- 0
	1	- 1

= (11001)

Octal

8	25	
	3	- 1

= (31)

Hexadecimal

16	25	
	1	- 9

= (19)

2. 73

Binary

2	73	
2	36	– 1
2	18	– 0
2	9	– 0
2	4	– 1
2	2	– 0
	1	– 0

= (1001001)

Octal

8	73	
8	9	– 1
	1	– 1

= (111)

Hexadecimal

16	73	
	4	– 9

= (49)

3. 255

Binary

2	255
2	127 - 1
2	63 - 1
2	31 - 1
2	15 - 1
2	7 - 1
2	3 - 1
	1 - 1

= (11111111)

Octal

8	255
8	31 - 7
	3 - 7

= (377)

Hexadecimal

16	255
	15 - 15

∴ 15 = F

= (FF)

4. 1023

Binary

2	1023
2	511 - 1
2	255 - 1
2	127 - 1
2	63 - 1
2	31 - 1
2	15 - 1
2	7 - 1
2	3 - 1
	1 - 1

= (1111111111)

Octal

8	1023
8	127 - 7
8	15 - 7
	1 - 7

= (1777)

Hexadecimal

16	1023
16	63 - 15
	3 - 15

∵ 15 = F

= (3FF)

Task 2: Binary Conversions

Convert the following binary numbers into **decimal and Hexadecimals:**

1. 101101

Decimal

$= 1\times2^5 + 0\times2^4 + 1\times2^3 + 1\times2^2 + 0\times2^1 + 1\times2^0$

$= 1\times32 + 0\times16 + 1\times8 + 1\times4 + 0\times2 + 1\times1$

$= 32 + 0 + 8 + 4 + 0 + 1$

$= 45$

Hexadecimal

Group from right in 4 bits → 0010 1101

0010 = 2

1101 = D

= 2D

2. 11110000

Decimal

$= 1\times2^7 + 1\times2^6 + 1\times2^5 + 1\times2^4 + 0\times2^3 + 0\times2^2 + 0\times2^1 + 0\times2^0$

$= 1\times128 + 1\times64 + 1\times32 + 1\times16 + 0\times8 + 0\times4 + 0\times2 + 0\times1$

$= 128 + 64 + 32 + 16$

$\therefore = 240$

Hexadecimal

Groups : 1111 0000

1111 = F

0000 = 0

= F0

3. 10011011

Decimal

$= 1\times2^7 + 0\times2^6 + 0\times2^5 + 1\times2^4 + 1\times2^3 + 0\times2^2 + 1\times2^1 + 1\times2^0$

$= 1\times128 + 0\times64 + 0\times32 + 1\times16 + 1\times8 + 0\times4 + 1\times2 + 1\times1$

$= 128 + 16 + 8 + 2 + 1$

$= 155$

Hexadecimal

Groups 1001 1011

1001 = 9

1011 = B

= 9B

4. 110011001

Decimal

$= 1\times2^8 + 1\times2^7 + 0\times2^6 + 0\times2^5 + 1\times2^4 + 1\times2^3 + 0\times2^2 + 0\times2^1 + 1\times2^0$

$= 1\times256 + 1\times128 + 0\times64 + 0\times32 + 1\times16 + 1\times8 + 0\times4 + 0\times2 + 1\times1$

$= 256 + 128 + 16 + 8 + 1$

$= 409$

CamScanner

Hexadecimal

Group 0001 1001 1001

0001 = 1

1001 = 9

1001 9

= 199

Task 3: Hexadecimal Conversions

Convert the following hexadecimal numbers into **Binary** and **Decimal**:

1. 2F

Binary

Each hex digit → 4 binary bits

2 = 0010

F = 1111

Combine 0010 1111

= (101111)

Decimal

$= 2 \times 16^1 + F \times 16^0$ $\because F = 15$

$= 2 \times 16 + 15 \times 1$

$= 32 + 15$

$= 47$

2. 7A

Binary

7 = 0111

A = 1010

combine 0111 1010

= (1111010)

Decimal

$= 7 \times 16^1 + A \times 16^0$

$= 7 \times 16 + 10 \times 1$ $\quad \because A = 10$

$= 112 + 10$

$= 122$

3. A3

Binary

A = 1010

3 = 0011

combine 1010 0011

= (10100011)

Decimal

$= A \times 16^1 + 3 \times 16^0$

$= 10 \times 16 + 3 \times 1$ $\quad \because A = 10$

$= 160 + 3$

$= 163$

4. FF

Binary

F = 1111

F = 1111

Combine → 1111 1111

= (11111111)

Decimal

$= F \times 16^1 + F \times 16^0$

$= 15 \times 16 + 15 \times 1$ ∵ F = 15

$= 240 + 15$

$= 255$

Task 4: Octal Conversions

Convert the following octal numbers into **Binary and Decimal**

1. 57

Binary

Each octal digits → 3 binary bits

5 = 101

7 = 111

combine 101 111

= 101111

Decimal

$= 5\times8^1 + 7\times8^0$

$= 5\times8 + 7\times1$

$= 40 + 7$

$= 47$

2. 124

Binary

1 = 001

2 = 010

4 = 100

combine 001 010 100

= 1010100

Decimal

$= 1\times8^2 + 2\times8^1 + 4\times8^0$

$= 1\times64 + 2\times8 + 4\times1$

$= 64 + 16 + 4$

$= 84$

3. 377

Binary

3 = 011

7 = 111

7 = 111

combine 011 111 111

= 11111111

Decimal

$$= 3\times8^2 + 7\times8^1 + 7\times8^0$$

$$= 3\times64 + 7\times8 + 7\times1$$

$$= 192 + 56 + 7$$

$$= 255$$

4. 645

Binary

6 = 110

4 = 100

5 = 101

combine 110 100 101

= 110100101

Decimal

$$= 6\times8^2 + 4\times8^1 + 5\times8^0$$

$$= 6\times64 + 4\times8 + 5\times1$$

$$= 384 + 32 + 5$$

$$= 421$$